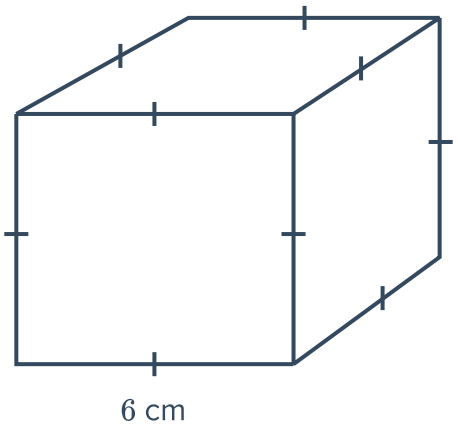
Surface area of prisms

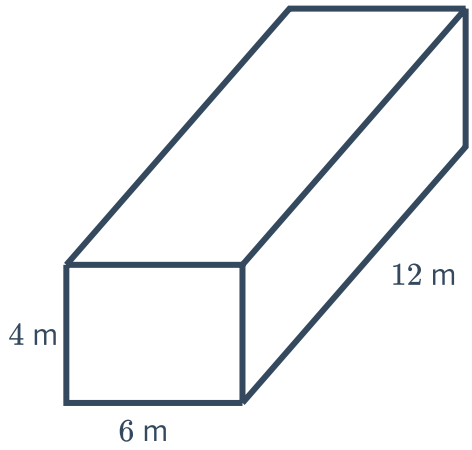
1 Find the surface area of the following cube with a side length of 6 cm:



2 Find the surface area of a cube with a side length of 4 cm.

3 Find the side length of a cube whose surface area is 8166 cm^2. Write your answer to Round decimal places.

4 Consider the following rectangular prism with length, width and height equal to 12 m, 6 m and 4 m respectively.
Find the surface area of the prism.

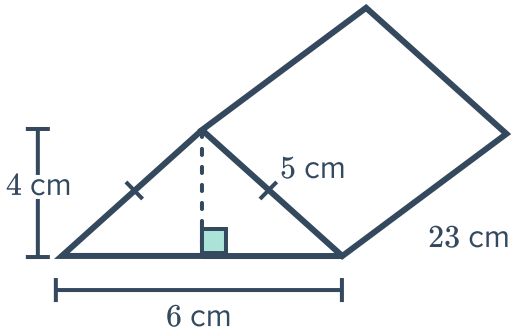


5 Find the surface area of a rectangular prism with dimensions: 4 cm by 9 cm by 8 cm.

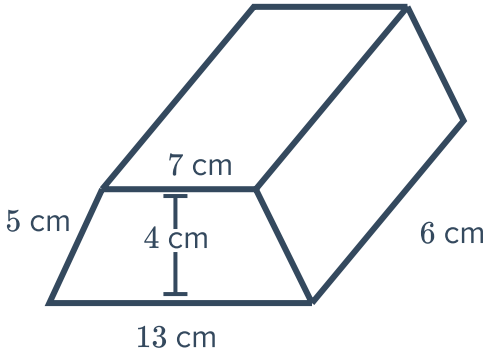
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6 Find the surface area of the prisms shown below:

a



b

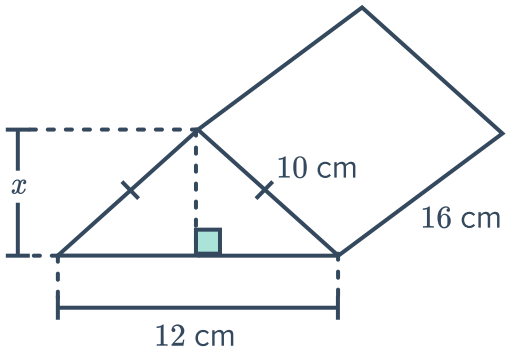


7 For each of the following triangular prisms:

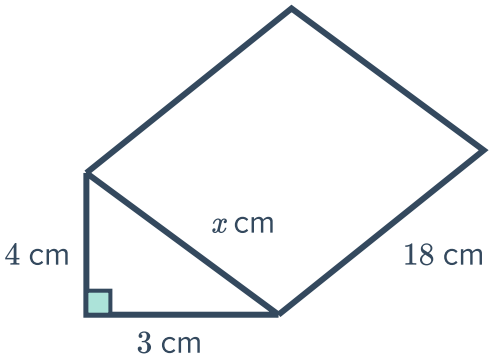
i Use Pythagoras' Theorem to find the value of x.

ii Find the surface area of the prism.

a



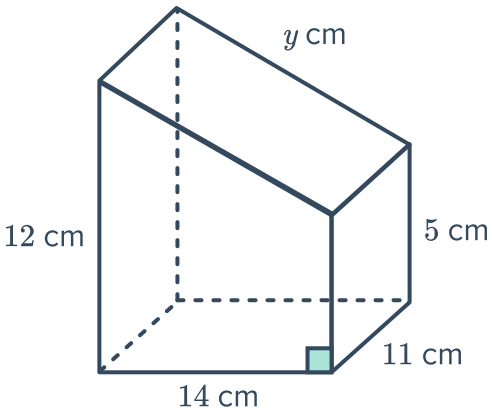
b



8 Consider the following trapezoidal prism:

a Find the value of y. Round your answer to two decimal places.

b Find the surface area of the trapezoidal prism. Round your answer to one decimal place.

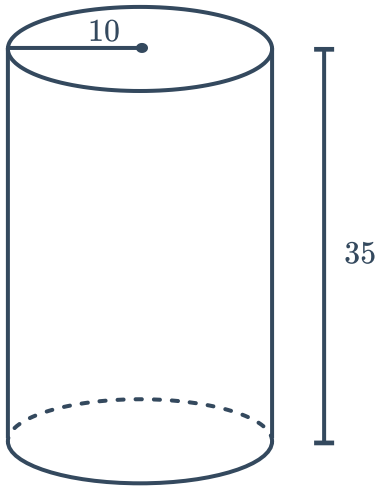


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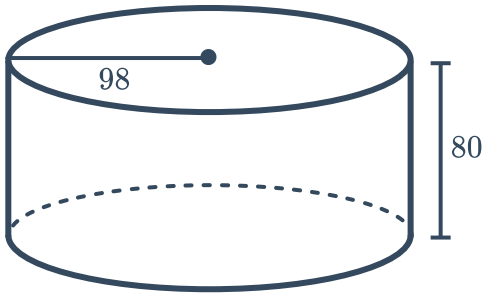
Surface area of cylinders

9 Find the surface area of the following cylinders. Round your answers to two decimal places.

a



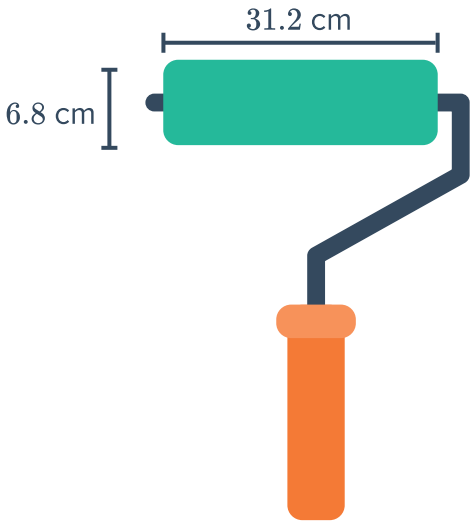
b



10 Find the exact surface area of a cylinder with diameter $6\ \text{cm}$ and height $21\ \text{cm}$, leaving your answer in terms of π.

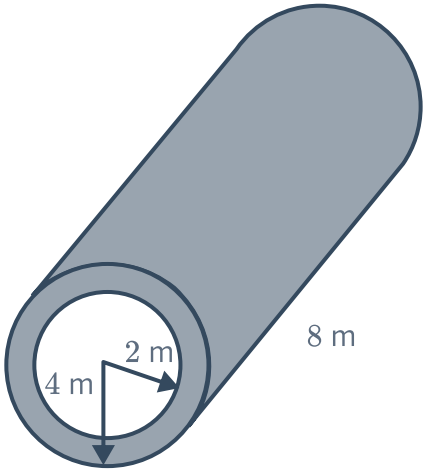
11 A cylindrical can of radius $7\ \text{cm}$ and height $10\ \text{cm}$ is open at one end. Find the external surface area of the can, correct to two decimal places.

12 A paint roller is cylindrical in shape. It has a diameter of $6.8\ \text{cm}$ and a width of $31.2\ \text{cm}$. Find the area painted by the roller when it makes one revolution, correct to the nearest $0.01\ \text{cm}^2$.



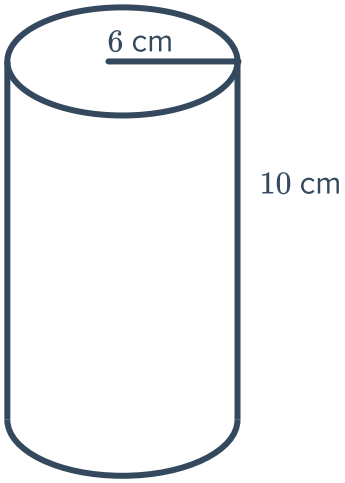
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13 Find the external surface area of a concrete pipe with outer radius $4\ \text{m}$, inner radius $2\ \text{m}$ and length $8\ \text{m}$, as shown in the figure. Round your answer to the nearest $0.01\ \text{m}^2$.



14 Find the lateral surface area of a cylinder with a radius of $9.2\,\text{m}$ and a height of $15.1\,\text{m}$. Round your answer to one decimal place.

15 Find the surface area of the cylinder shown. Round your answer to the two decimal places.

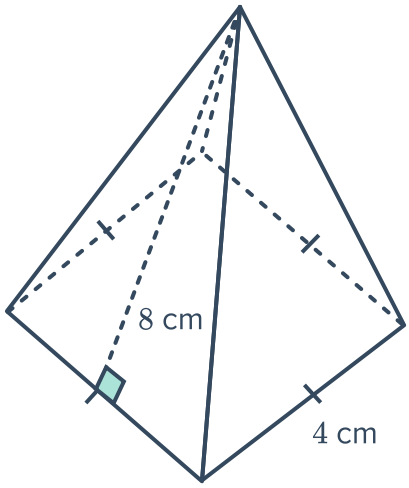


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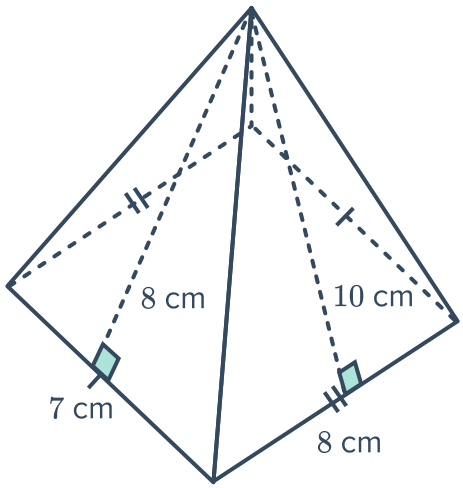
Surface area of pyramids

16 Find the surface area of the following pyramids:

a



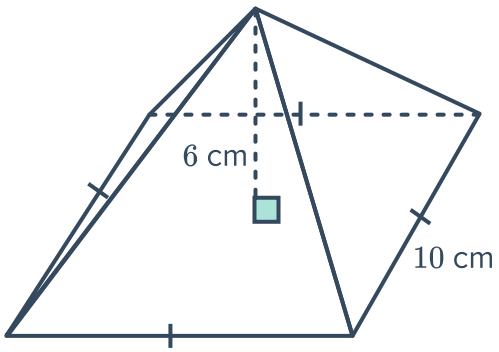
b



17 Consider the following square pyramid:

a Find the length of the slant height. Round your answer to two decimal places.

b Hence, find the surface area of the square pyramid. Round your answer to one decimal place.



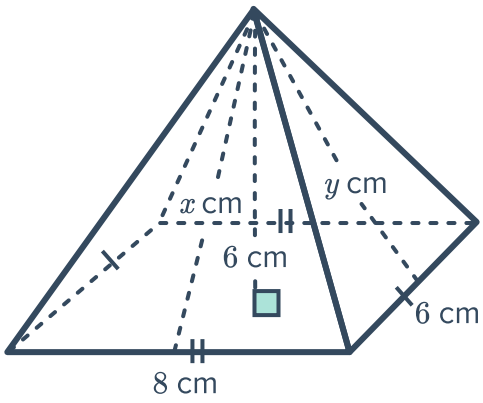
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18 Consider the rectangular pyramid with rectangular base of dimensions 6 cm by 8 cm and perpendicular height 6 cm:

a Find the length of the slant height, x cm, correct to two decimal places.

b Find the length of the second slant height, y cm, correct to two decimal places.

c Find the surface area of the pyramid. Round your answer to one decimal place.



19 Find the surface area of a square pyramid with base edge 10 cm and perpendicular height 5 cm. Round your answer to two decimal places.

20 Find the surface area of a rectangular pyramid with base 10 cm by 5 cm and perpendicular height 4 cm. Round your answer to one decimal place.

21 A square pyramid has a surface area of $480\ \text{cm}^2$. The area of its square base is $36\ \text{cm}^2$.

a Find the area of each triangular face.

b Find the slant height of each triangular face, correct to two decimal places.

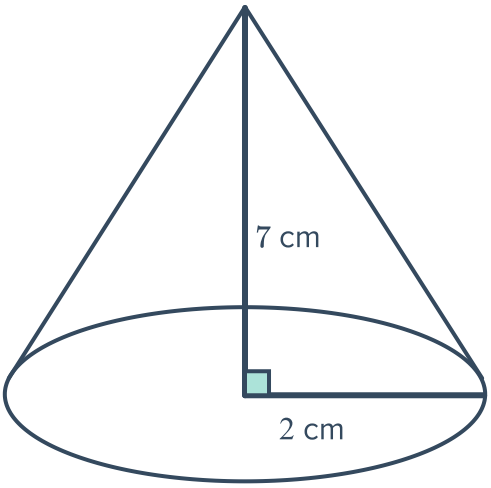
c Find the perpendicular height of the pyramid, correct to one decimal place.

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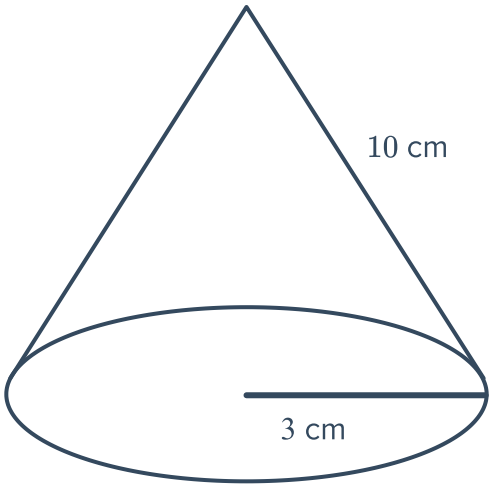
Surface area of cones and spheres

22 Find the surface area of the following cones, round your answers to two decimal places:

a



b



23 A cone has a surface area of $560\ \text{cm}^2$ and the radius of its base is $6\ \text{cm}$.

a Find the slant height of the cone. Round your answer to two decimal places.

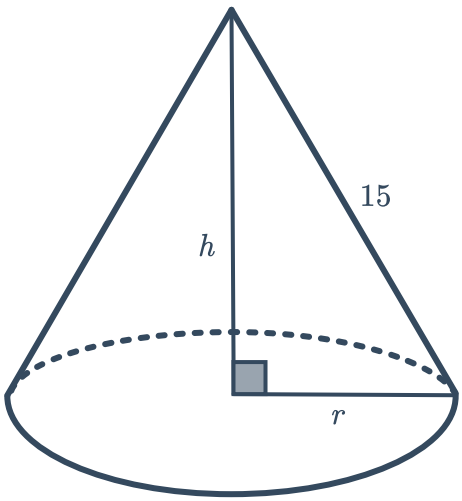
b Hence, find the perpendicular height of the cone. Round your answer to one decimal place.

24 The curved area of a cone is $380\ \text{cm}^2$ and the slant height is $15\ \text{cm}$:

a Find the radius of the cone, correct to two decimal places.

b Find the perpendicular height of the cone, correct to one decimal place.

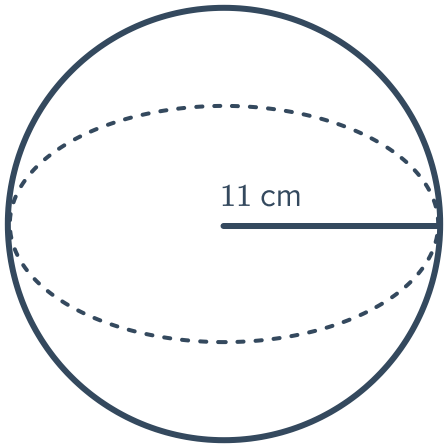
c Find the total surface area of the cone, correct to the nearest whole number.



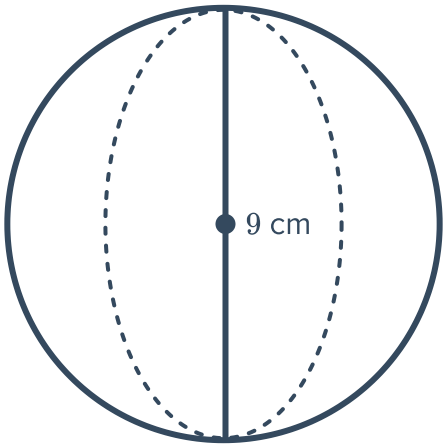
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25 Find the surface area of the following spheres. Round your answers to two decimal places.

a



b

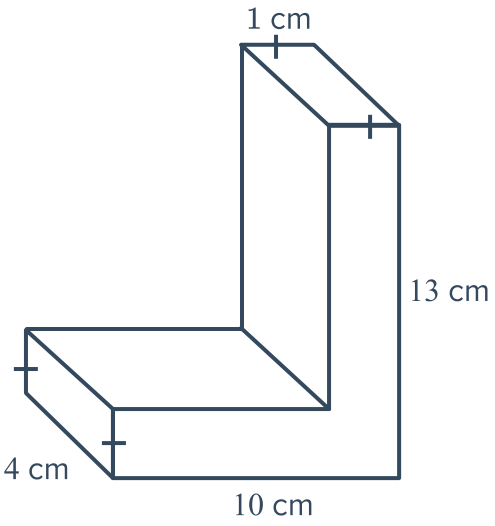


26 A sphere has a surface area of $450\ \text{cm}^2$. Find the radius of the sphere. Round your answer to two decimal places.

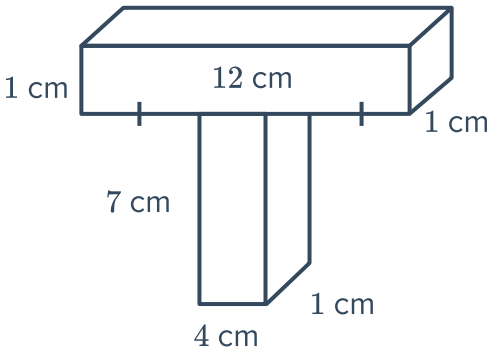
Applications

27 Find the surface area of the following composite solids. Round your answer to two decimal places where necessary.

a

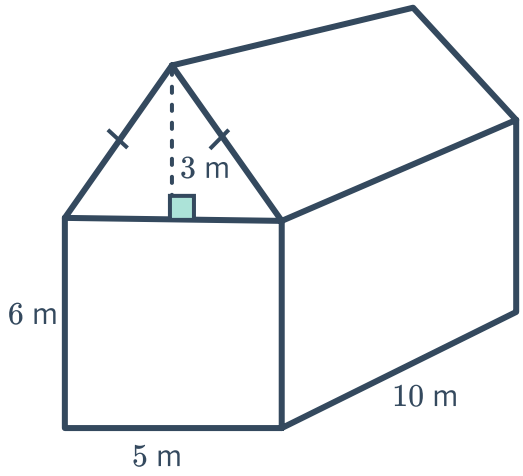


b

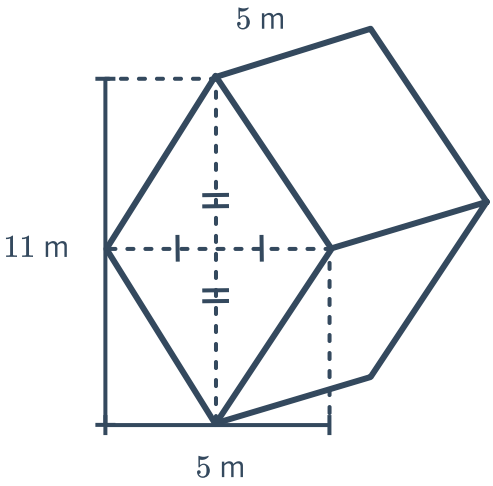


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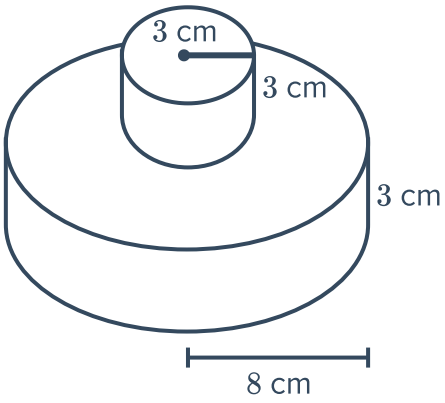
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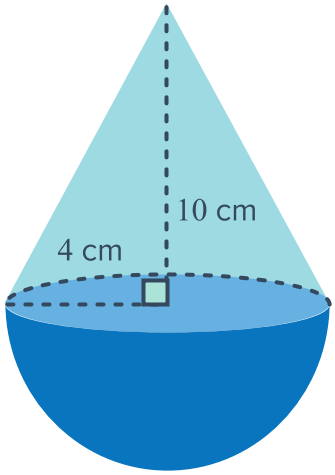
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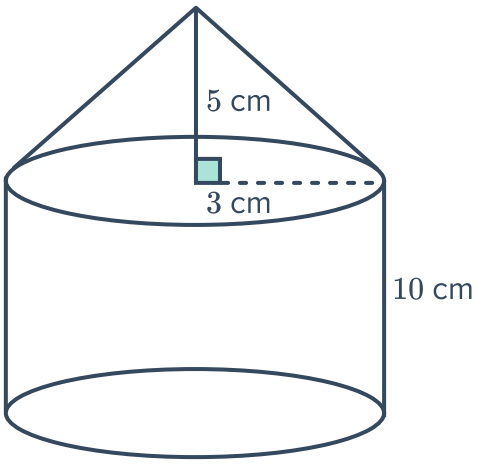
e



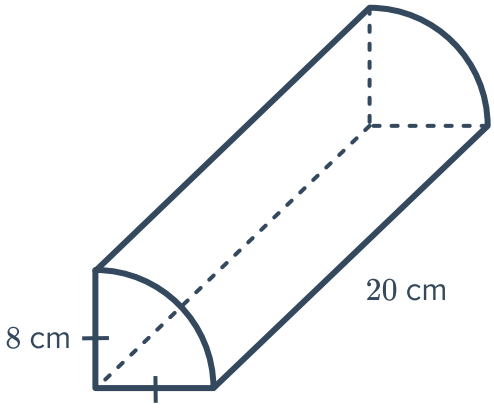
f



g

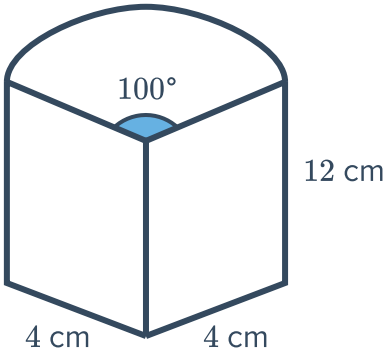


h

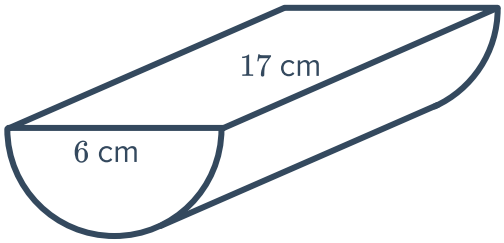


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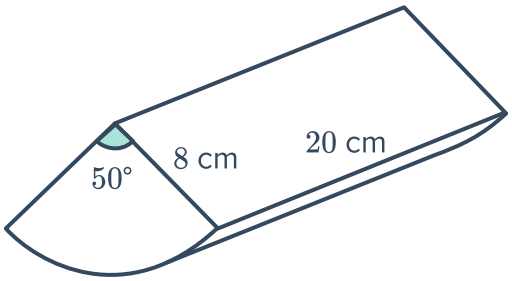
i



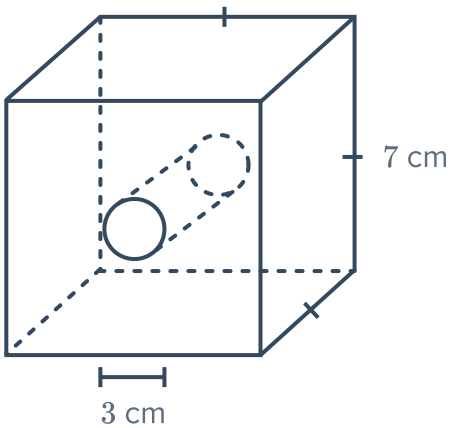
j



k

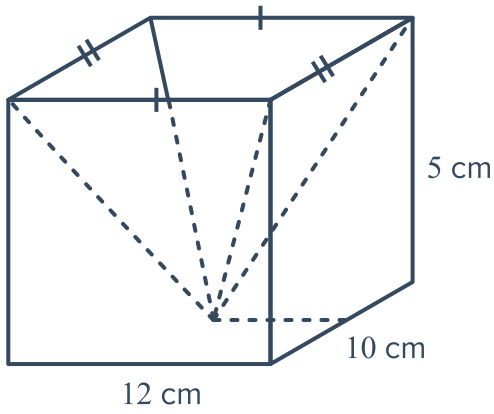


28 A hole of diameter 3 cm has been drilled through the centre of a solid cube of side length 7 cm. The resulting solid is to be painted to protect it from rust. Find the total surface area of the solid to be painted correct to the nearest cm^2

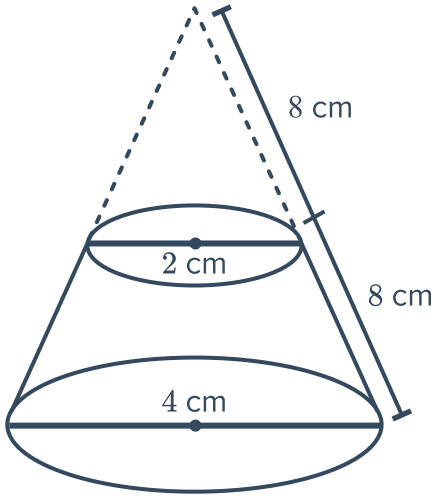


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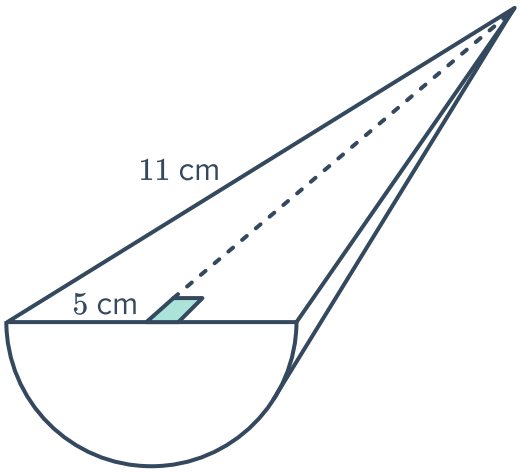
29 A pyramid has been removed from a rectangular prism, as shown in the figure. Find the surface area of the composite solid.



30 The top of a solid cone was sawed off to form the solid shown. Find the surface area of the solid formed correct to two decimal places.



31 The diagram shows a cone of diameter 5 cm and slant height 11 cm sliced in half. Find the surface area of the solid. Round your answer to two decimal places.

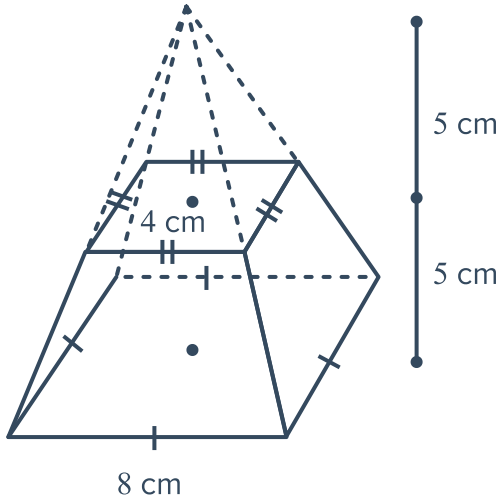


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32 A small square pyramid of height 5 cm was removed from the top of a large square pyramid of height 10cm to form the solid shown:

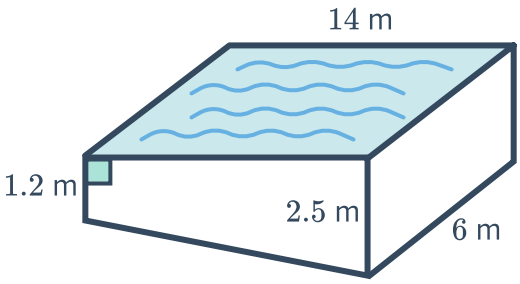
a Find the length of the slant height of the sides of the new solid. Round your answer to two decimal places.

b Find the surface area of the solid formed. Round your answer to one decimal place.

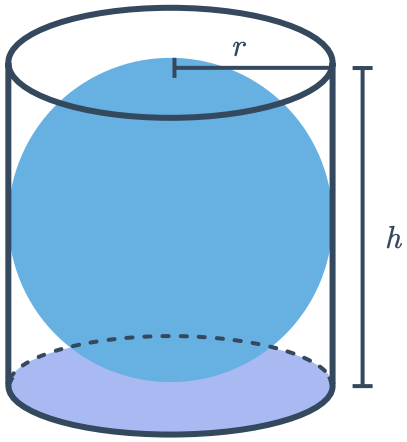


33 If a spherical ball with a radius of 3.7 cm fits exactly inside a cylinder, what is the surface area of the cylinder? Round your answer to one decimal place.

34 A swimming pool has the shape of a trapezoidal prism 14 m long and 6 m wide. The depth of the water ranges from 1.2 m to 2.5 m, as shown in the diagram:
The walls and base of the pool are going to be tiled. Calculate the total area inside the pool that is to be tiled. Round your answer to one decimal place.

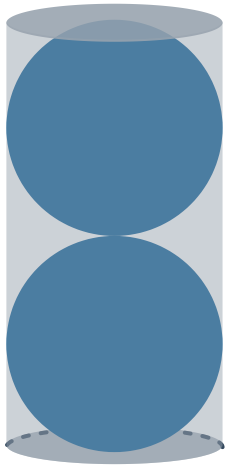


35 If a cylinder with a radius of $r = 3.1$ cm and a height of $h = 6.2$ cm contains a ball that fits exactly inside it, what is the surface area of the ball? Round your answer to one decimal place.



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36 If two identical spherical balls with radii of $1.4\ \text{m}$ fit exactly inside a cylinder, what is the surface area of the closed cylinder? Round your answer to one decimal place.



37 A cylinder with a radius of $2.5\ \text{cm}$ and a height of $10\ \text{cm}$ contains two identical spherical balls that fit exactly inside it. What is the surface area of each ball? Round your answer to one decimal place.

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